

Lab-7

1. Apply apriori algorithm on given dataset, Calculate support and confidence then find the association.

TID	Items	Minimum Support = 2
100	1 3 4	
200	2 3 5	
300	1 2 3 5	
400	2 5	

Scan D $\rightarrow$ C_1	Itemset	Min. Sup	$\rightarrow$ L_1	Itemset	Min Sup
	{1}	2		{1}	2
	{2}	3		{2}	3
	{3}	3		{3}	3
	{4}	1		{5}	3
	{5}	3			

C_2	Itemset		C_2	Itemset	Min Sup
	{1 2}	Scan D		{1 2}	1
	{1 3}	$\rightarrow$		{1 3}	2
	{1 5}			{1 5}	1
	{2 3}			{2 3}	2
	{2 5}			{2 5}	3
	{3 5}			{3 5}	2

$\rightarrow L_2$	Itemset	Min Sup	scan $\rightarrow L_3$	Item Set	Min Sup
	{1 3}	2	$\rightarrow$	{1 2 3}	1
	{2 3}	2		{1 3 5}	1
	{2 5}	3		{2 3 5}	2
	{3 5}	2			

Rule Generation :-

Association Rule	Support	Confidence	Confidence (%)
$2\ 3 \rightarrow 5$	2	$2/2 = 1$	100%
$3\ 5 \rightarrow 2$	2	$2/2 = 1$	100%
$2\ 5 \rightarrow 3$	2	$2/3 = 0.66$	66%
$2 \rightarrow 3\ 5$	2	$2/3 = 0.66$	66%
$3 \rightarrow 2\ 5$	2	$2/3 = 0.66$	66%
$5 \rightarrow 2\ 3$	2	$2/3 = 0.66$	66%

2.	TID	Items	
	1	Bread, Milk	Minimum
	2	Bread, Diaper, Beer, Eggs	Support = 2
	3	Milk, Diaper, Beer, Cola	
	4	Milk, Diaper, Beer, Cola	
	5	Bread, Milk, Diaper, Cola	

Here Let, Bread = 1, Milk = 2, Diaper = 3, Beer = 4, Eggs = 5, Cola = 6

TID	Items	C ₁ Items	Min. Sup
1	1 2	scan D {1}	3
2	1 3 4 5	→ {2}	4
3	2 3 4 6	{3}	4
4	2 3 4 6	{4}	3
5	1 2 3 6	{5}	1
		{6}	3

L ₁	Item Set	Min. Sup	C ₂	Item Set	Min. Sup
	{1}	3	scan D	{1, 2}	2
	{2}	4	→	{1, 3}	2
	{3}	4		{1, 4}	1
	{4}	3		{1, 6}	1
	{6}	3		{2, 3}	3
				{2, 4}	2
				{2, 6}	3
				{3, 4}	3
				{3, 6}	3
				{4, 6}	2

L_2	ItemSet	Min. Sup	C_2	ItemSet	Min. Sup
	{1, 2}	2		{1, 2, 3}	1
	{1, 3}	2		{1, 2, 4}	0
	{2, 3}	3	scan	{1, 2, 6}	1
	{2, 4}	2	$\rightarrow$	{1, 3, 4}	1
	{2, 6}	2		{1, 3, 6}	1
	{3, 4}	3		{2, 3, 4}	2
	{3, 6}	3		{2, 3, 6}	3
	{4, 6}	2		{2, 4, 6}	2
				{3, 4, 6}	2

L_3	ItemSet	Min. Sup	C_3	ItemSet	Min. Sup
	{2, 3, 4}	2	scan $\rightarrow$	{2, 3, 4, 6}	2
	{2, 3, 6}	3			
	{2, 4, 6}	2			
	{3, 4, 6}	2			

Rule Generation:-

Association Rule	Support	Confidence	Confidence(%)
$2 \rightarrow 3 \ 4 \ 6$	2	$2/4 = 0.5$	50%
$3 \rightarrow 2 \ 4 \ 6$	2	$2/4 = 0.5$	50%
$4 \rightarrow 2 \ 3 \ 6$	2	$2/3 = 0.66$	66%
$6 \rightarrow 2 \ 3 \ 4$	2	$2/3 = 0.66$	66%
$2 \ 3 \rightarrow 4 \ 6$	2	$2/3 = 0.66$	66%
$2 \ 4 \rightarrow 3 \ 6$	2	$2/2 = 1$	100%
$2 \ 6 \rightarrow 3 \ 4$	2	$2/3 = 0.66$	66%
$3 \ 4 \rightarrow 2 \ 6$	2	$2/3 = 0.66$	66%
$3 \ 6 \rightarrow 2 \ 4$	2	$2/3 = 0.66$	66%

